Pranali Sandeep Hagare

Artificial Intelligence and Data Science
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Education

- **Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering & Technology** *June 2024*
  Bachelor of Engineering in Artificial Intelligence & Data Science - CGPA : 8.19 Baramati, Pune

Experience

- **Test Yantra** *November 2025-April 2026*
  Associate Software Engineer
  - Built Python scripts for data preprocessing and transformation to support ML training pipelines
  - Performed EDA using Pandas and NumPy to identify patterns in large application datasets
  - Worked on structured data workflows supporting AI model development and evaluation
- **Oasis Infobyte [certificate]** *February 2023 - March 2023*
  Data Science Intern Remote
  - Managed diverse case studies involving image and text datasets, performing tasks such as exploratory data analysis (EDA), data cleaning, transformation, model building, testing, and deriving insights.
  - Utilized data visualization techniques to communicate findings and insights effectively, contributing to informed decision-making processes based on data-driven analysis.

Projects

- **Indian Sign Language Interpreter** *December 2023 -July 2024*
  Tools: [TensorFlow, OpenCV, Mediapipe, NumPy, scikit-learn, Matplotlib] []
  - **Technologies Used:** Holistic Model for Pose and Hand Detection, LSTM Neural Network, Data Preprocessing, Real-Time Gesture Recognition
  - Developed an LSTM-based gesture recognition system with **96.25% accuracy**, classifying 8 gestures using **Mediapipe** for real-time keypoint extraction. Collected and processed over **50** video sequences per gesture, training the model with advanced **LSTM architecture** and **RMSprop optimizer**. Achieved an average **Dice Score of 94.82%** and visualized real-time predictions with **OpenCV** for enhanced user interaction.
- **Agriculture Production Optimization Engine:** *December 2022 - July 2023*
  Tools: [NumPy, Pandas, Matplotlib, Seaborn, Ipywidgets] []
  - **Technologies Used:** Logistic Regression, KMeans Clustering, Data Visualization, Elbow Method
  - Applied **KMeans** Clustering to group **22** different crops based on soil and environmental features, increasing recommendation precision by **15%**. Achieved **88% accuracy** in crop prediction using **Logistic Regression** and Reduced decision-making time by **20%** with real-time nutrient and weather insights using **Ipywidgets**. Identified key patterns in crops requiring over 120 kg nitrogen, 100 kg phosphorous, and 200 mm rainfall using data analysis and visualization.

Skills

- **Programming Languages**: Python, SQL
- **Python Libraries:**:NumPy, Pandas, Matplotlib, Seaborn
- **Data Science / Machine Learning / Deep Learning**: Python, Data Visualization, EDA, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Ensemble Learning, Clustering Algorithms, ANN, CNN, RNN, LSTM, Feature Engineering, Feature selection,Object-Oriented Programming (OOP),NLP

Publication

- **"Empowering Deaf by Translating Emergency Signs in Indian Sign Language to Text"**, Copyright Office, Government of India. Reg No: L-137992/2023 **Certificate**
- **"Empowering Deaf with Indian Sign Language Interpreter using Deep Learning"**, 2024 MIT Art, Design and Technology School of Computing International Conference (MITADTSoCiCon), Pune, India, 2024, pp. 1-6, doi: 10.1109/MITADTSoCiCon60330.2024.10575064. **Certification**